

Pressure in Solids and Liquids — Paper B

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a **unit**. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: **pressure = force ÷ area**, and in a liquid **pressure = density × g × depth**.
- Take **g = 10 N/kg**, and remember that **1 m² = 10 000 cm²**.

SECTION 1 — REAL CONTACT AREAS

- 1.** Aarav weighs 600 N. Standing on both feet, his shoes touch the floor over a total area of 0.03 m².

(a) Calculate the pressure he exerts on the floor. [2]

(b) He lifts one foot. Calculate the new pressure on the floor. [2]

(c) State what happened to the force he exerts on the floor. [1]

- 2.** A metal crate of weight 900 N stands on four small feet. Each foot has an area of 25 cm².

(a) Calculate the total contact area, in m². [2]

(b) Calculate the pressure on the floor. [2]

(c) A board of area 0.3 m^2 is slid underneath. Calculate the new pressure. [1]

SECTION 2 – PRESSURE IN LIQUIDS

3. Calculate the pressure due to the liquid in each case.

(a) Water of density 1000 kg/m^3 at a depth of 3 m. [2]

(b) Seawater of density 1030 kg/m^3 at a depth of 20 m. [2]

4. A tall narrow jar and a wide tank are both filled with water to a depth of 0.5 m. A student says the pressure at the bottom of the wide tank must be greater, because it holds more water. Explain why he is wrong, and calculate the pressure at the bottom of each. [4]

5. Explain why a dam is built much thicker at the bottom than at the top. [3]

6. A sealed can is filled with water. Three small holes are made in its side, at different heights. Describe what you would see, and explain both features of what you have described. [4]

7. State what is meant by **atmospheric pressure**, and explain why you do not normally notice it. [3]

8. A diver is 12 m below the surface of a freshwater lake. Taking atmospheric pressure as 100 000 Pa, calculate the total pressure acting on her. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.